MODEL QUESTION PAPER 1

S3 INSTRUMENTATION ENGINEERING

SUBJECT: SENSORS AND TRANSDUCERS

Time: 3 Hours Max. Marks: 75

I: Part – A: Answer all the following questions in one word or sentence. (9 x 1 = 9 Marks)

1. Define the transducer.
2. List two applications of resistive transducers.
3. Name two types of inductive transducers.
4. Write the equation for voltage sensitivity of piezoelectric transducers.
5. Name a photo conductive transducer.
6. Write one application of Hall effect transducer.
7. What is the use transmitters.
8. List one application of nano sensors.
9. Write one classification of transducers.

II: Part - B: Answer any eight questions. Each question carries 3 marks (8 x 3 = 24 Marks)

1. With examples describe two classifications of transducers.
2. Explain loading effect.
3. Describe the basic principle of eddy current transducer. .
4. Draw the equivalent circuit of piezoelectric transducer.
5. Describe the working principle of photo diode
6. Explain the level measurement using ultrasonic transducer.
7. Illustrate one application of nano sensors.
8. Explain the difference between conventional sensors and smart sensors.
9. Describe two types of strain gauges
10. Write short notes on industrial transmitters

Part - C: Answer all questions. Each question carries 7 marks. (6 x 7 = 42 Marks)

III. Derive the expression for loading error in potentiometers .

OR

IV. Illustrate construction, working and characteristics of light dependent resistors (LDR).

V. With necessary figures explain construction and working of liner

Variable differential transformers (LVDT).

OR

VI. Explain the pressure measurement using piezoelectric transducer.

VII. With necessary figures explain construction and working of photomultiplier tube.

OR

VIII. Explain construction and working of Hall effect transducers.

IX. With necessary figure explain working of smart transmitters.

OR

X. With necessary figures explain construction and working of capacitive type differential pressure transmitters.

XI. With necessary figures explain any one application of potentiometer.

OR

XII. Derive the expression for gauge factor of strain gauge.

XIII. With necessary figures explain any one application of LVDT

OR

XIV. With necessary figures explain any one application of capacitive transducers

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Question Wise Analysis

Course: **SENSORS AND TRANSDUCERS** – Question -1

Q. No	Module Outcome	Cognitive Level	Marks	Time(Min.)
I.1	M1.01	U	1	2
I.2	M1.03	U	1	2
I.3	M2.01	R	1	2
I.4	M2.04	U	1	2
I.5	M3.01	U	1	2
I.6	M3.03	U	1	2
I.7	M4.01	U	1	2
I.8	M4.02	U	1	2
I.9	M1.01	U	1	2
II.1	M1.01	U	3	8
II.2	M1.02	U	3	8
II.3	M2.02	U	3	8
II.4	M2.04	U	3	8
II.5	M3.01	U	3	8
II.6	M3.02	U	3	8
II.7	M4.02	U	3	8
II.8	M4.02	U	3	8
II.9	M1.02	U	3	8
II.10	M4.03	U	3	8
III	M1.02	A	7	16
IV	M1.02	U	7	16
V	M2.02	U	7	16
VI	M2.04	U	7	16
VII	M3.01	U	7	16
VIII	M3.03	U	7	16
IX	M4.02	U	7	16
X	M4.03	U	7	16
XI	M1.03	U	7	16
XII	M1.02	A	7	16
XIII	M2.02	U	7	16
XIV	M2.03	U	7	16

Course : **SENSORS AND TRANSDUCERS –QP-1**

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Mark Distribution

Module	hr / Module	Marks/Module (h_i /$\sum H_i$) * 123 (±5%)	Type of Questions							
			Part A		Part B		Part C		Total	
			No of Questions	Marks	No of Questions	Marks	No of Questions	Marks	No of Questions	Marks
1	13	35.5 ± 6	3	3	3	9	4	28	9	39
2	11	30 ± 6	2	2	2	6	4	28	8	36
3	10	27 ± 6	2	2	2	6	2	14	6	22
4	11	30 ± 6	2	2	3	9	2	14	8	26
Total	45	123	9	9	10	30	12	84	31	123

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Cognitive Level Mark Distribution

Cognitive Level	Marks	% of Marks
Remembering	1	1
Understanding	108	88
Applying	14	11
Analysing	0	0
Evaluating	0	0
Creating	0	0
Total	123	100

MODEL QUESTION PAPER 2

S3 INSTRUMENTATION ENGINEERING

SUBJECT: SENSORS AND TRANSDUCERS

Time: 3 Hours Max. Marks: 75

I: Part – A: Answer all the following questions in one word or sentence. (9 x 1 = 9 Marks)

1. Give an example each for primary and secondary transducer.
2. List one applications of strain gauge.
3. Name two types of capacitive transducers.
4. Write the equation for charge sensitivity of piezoelectric transducers.
5. List any two photosensitive materials.
6. Write one application of photoelectric transducer.
7. What are the types of industrial transmitters?
8. What do you mean by nano sensor?
9. Write one classification of transmitters.

II: Part - B: Answer any eight questions. Each question carries 3 marks. (8 x 3 = 24 Marks)

1. Distinguish between sensors and transducers.
2. Write down the expression for loading error in potentiometer.
3. Describe the basic principle of Inductive transducer.
4. Draw the equivalent circuit of piezoelectric transducer.
5. Explain current measurement using Hall effect transducer.
6. Draw the picture of the photo multiplier tube.
7. Discuss on the concept of standard electrical transmission.
8. Explain the basics of nano sensors with an example.
9. Compare RTD and thermistor.
10. List the advantages of smart transmitters over conventional transmitters

Part - C: Answer all questions. Each question carries 7 marks. (6 x 7 = 42 Marks)

III. Derive the expression for gauge factor of a strain gauge.

OR

IV. Explain how displacement is measured using potentiometer.

V. With necessary figures explain construction and working of liner variable differential transformers (LVDT).

OR

VI. Explain the level measurement using capacitive transducer.

VII. With necessary figures explain one application of ultrasonic transducers.

OR

VIII. Explain construction and working of photovoltaic cell.

IX. With necessary figures explain construction and working of capacitive type differential pressure transmitters.

OR

X. With necessary figure, explain smart sensors.

XI. Explain any three types of strain gauges.

OR

XII. With necessary examples, explain:

(i) active and passive transducers.

(ii) primary and secondary transducers.

XIII. With necessary figures explain any two applications of piezo electric transducer.

OR

XIV. With necessary figures explain pressure measurement using LVDT.

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Question Wise Analysis

Course: **SENSORS AND TRANSDUCERS** – Question -2

Q. No	Module Outcome	Cognitive Level	Marks	Time(Min.)
I.1	M1.01	U	1	2
I.2	M1.03	U	1	2
I.3	M2.03	R	1	2
I.4	M2.04	U	1	2
I.5	M3.01	U	1	2
I.6	M3.01	U	1	2
I.7	M4.03	U	1	2
I.8	M4.02	U	1	2
I.9	M4.03	U	1	2
II.1	M1.01	U	3	8
II.2	M1.02	U	3	8
II.3	M2.01	U	3	8
II.4	M2.04	U	3	8
II.5	M3.03	U	3	8
II.6	M3.01	U	3	8
II.7	M4.01	U	3	8
II.8	M4.02	U	3	8
II.9	M1.03	U	3	8
II.10	M4.02	U	3	8
III	M1.02	A	7	16
IV	M1.03	U	7	16
V	M2.02	U	7	16
VI	M2.03	U	7	16
VII	M3.02	U	7	16
VIII	M3.01	U	7	16
IX	M4.03	U	7	16
X	M4.02	U	7	16
XI	M1.02	U	7	16
XII	M1.01	U	7	16
XIII	M2.04	U	7	16
XIV	M2.02	U	7	16

Course : **SENSORS AND TRANSDUCERS –QP-1**

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Mark Distribution

Module	hr / Module	Marks/Module ($h_i / \sum H_i$) * 123 (±5%)	Type of Questions							
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			No of Questions	Marks	No of Questions	Marks	No of Questions	Marks	No of Questions	Marks
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Total	45	123	9	9	10	30	12	84	31	123

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Cognitive Level Mark Distribution

Cognitive Level	Marks	% of Marks
Remembering	1	1
Understanding	115	93
Applying	7	6
Analysing	0	0
Evaluating	0	0
Creating	0	0
Total	123	100